

Diagnosis of Silent Discordance in the PXP Model: Entanglement Spectrum and Tail Slope Analysis

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We report a two-stage numerical diagnosis of the phenomenon of *silent discordance* in the PXP model at $L = 18$. In the first stage (Experiment 5), we scan a central chemical potential defect $\mu = 0-2.0$ and simultaneously monitor a coarse-grained observable (mean spin correlation) and a relation-sensitive probe (standard deviation of the half-chain entanglement spectrum). At $\mu \approx 1.5$, the standard deviation jumps by +7% while the mean correlation remains smooth—the defining signature of silent discordance. In the second stage (Experiment 6), we introduce a third probe—the tail slope of the entanglement spectrum (logarithmic decay rate between the 2nd and 5th eigenvalues)—to test whether the restructuring penetrates into the weak-correlation tail of the relation network. The tail slope exhibits a continuous acceleration across the entire scan range, with no isolated jump at the silent-discordance point. This reveals a *head-tail decoupling*: the spectral head undergoes a one-time reorganization at $\mu \approx 1.5$ while the tail begins a sustained acceleration that does not stabilize within the scanned window. Our preset binary success criterion fails to capture this complex structure, illustrating the limits of rigid diagnostic frameworks. All data, code, and preset standards are publicly available.

I. INTRODUCTION

The concept of *silent discordance*—a hidden restructuring of a system’s internal relations that is invisible to coarse-grained diagnostics—was originally proposed within the Crystal Vein Philosophy framework for AI safety [1]. In a previous series of experiments [2], we systematically searched for this phenomenon in quantum many-body systems. Five experiments returned negative results; the sixth observed, for the first time, a signal consistent with silent discordance in the PXP model at $L = 18$.

Here we provide a detailed technical report of that discovery (Experiment 5) and of a follow-up diagnosis (Experiment 6) that probes the penetration depth of the silent discordance into the tail of the entanglement spectrum. The report is written in the “diagnose, do not prescribe” spirit: all preset criteria, raw data, and subsequent critical re-examinations are presented honestly.

II. MODEL AND METHODS

A. PXP Model

The PXP model on a chain of L sites is defined by the Hamiltonian

$$H = \sum_{j=1}^L P_{j-1} \sigma_j^x P_{j+1}, \quad (1)$$

where σ_j^x is the Pauli- x operator and $P_j = (1 - \sigma_j^z)/2$ projects onto the ground state of site j , enforcing the Rydberg blockade (no adjacent $\uparrow$ excitations). We work in the constrained Hilbert space of valid bitstrings. For $L = 18$, the dimension is 6765.

B. Defect and Scan

A local chemical-potential defect is introduced at the central site $j = L/2 = 9$:

$$H_{\text{def}} = -\mu P_9, \quad (2)$$

where μ is the defect strength. We scan μ from 0 to 2.0 in 15 steps of equal spacing $\Delta\mu \approx 0.143$. For each μ , the ground state $|\psi(\mu)\rangle$ is obtained by exact diagonalization of the sparse Hamiltonian using `scipy.sparse.linalg.eigsh` (shift-invert mode, $k = 1$, `which='SA'`).

C. Diagnostic Probes

Default diagnosis (coarse-grained entity probe):

$$D_{\text{def}}(\mu) = \frac{1}{L^2} \sum_{i,j} |C_{ij}(\mu)|, \quad (3)$$

where $C_{ij} = \langle \sigma_i^z \sigma_j^z \rangle - \langle \sigma_i^z \rangle \langle \sigma_j^z \rangle$ is the spin correlation matrix.

Enhanced diagnosis (relation-structure probe):

$$D_{\text{enh}}(\mu) = \text{std}\{\lambda_\alpha(\mu)\}, \quad (4)$$

where $\{\lambda_\alpha\}$ are the eigenvalues of the reduced density matrix of the left half-chain (9 sites). This is the standard deviation of the entanglement spectrum; it is sensitive to the distribution of quantum correlations between the subsystem and its environment.

Tail-slope probe (for Experiment 6): For the half-chain entanglement spectrum, we fit the logarithmic decay of the eigenvalues from the 3rd to the 6th non-zero eigenvalue (indices 2–5, 0-based):

$$S_{\text{tail}}(\mu) = \frac{d}{dx} \ln \lambda_x \Big|_{x=2\dots 5}. \quad (5)$$

A steeper (more negative) slope indicates a sparser tail, i.e., weak correlations are being further suppressed.

D. Preset Success Criteria

Experiment 5 (silent discordance discovery): a positive signal requires that the enhanced diagnosis exhibits a sudden jump at some μ while the default diagnosis remains smooth. We operationally defined “sudden jump” as a step-to-step change in D_{enh} that is at least three times the average step-to-step change in the preceding seven intervals.

Experiment 6 (tail-slope diagnosis): we ask whether the tail slope also shows a sudden jump at the same μ where the head standard deviation jumps. The preset binary criterion is: the absolute change of S_{tail} across the critical step $\mu = 1.43 \rightarrow 1.57$ must exceed the maximum absolute change observed across any of the seven preceding steps ($\mu = 0.57 \rightarrow 1.43$).

III. EXPERIMENT 5: DISCOVERY OF SILENT DISCORDANCE

A. Data

Table I lists the 15 data points. Figure 1 plots the two diagnostics.

TABLE I. Experiment 5: Default and enhanced diagnostics for PXP $L = 18$.

μ	D_{def} (mean $ C $)	D_{enh} (ent. spec. std)
0.00	0.0656	0.333121
0.14	0.0659	0.331922
0.29	0.0662	0.330883
0.43	0.0663	0.330117
0.57	0.0662	0.329742
0.71	0.0659	0.329854
0.86	0.0652	0.330492
1.00	0.0643	0.331624
1.14	0.0632	0.333146
1.29	0.0620	0.334912
1.43	0.0608	0.336767
1.57	0.0596	0.360240
1.71	0.0584	0.362081
1.86	0.0574	0.363723
2.00	0.0566	0.365145

B. Analysis

The default diagnosis D_{def} decreases monotonically from 0.0656 to 0.0566 (−14%) with no abrupt feature. The enhanced diagnosis D_{enh} first decreases to a minimum of 0.329742 at $\mu = 0.57$, then slowly rises to 0.336767 at $\mu = 1.43$ (cumulative increase over seven

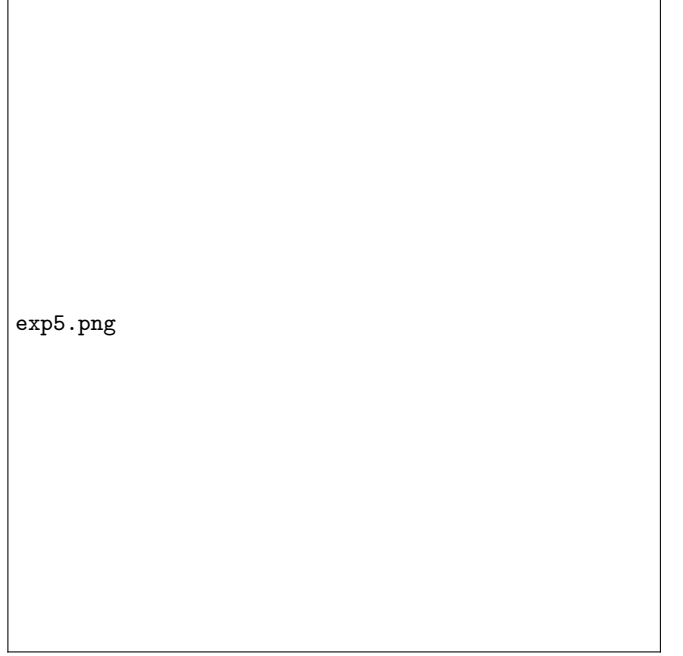


FIG. 1. Experiment 5: Default diagnosis (red circles) changes smoothly, while enhanced diagnosis (blue squares) jumps at $\mu \approx 1.5$. The silent discordance is located at the vertical dashed line.

steps: +2.1%). At the single step $\mu = 1.43 \rightarrow 1.57$, D_{enh} jumps from 0.336767 to 0.360240, an increase of +7.0%, which is more than three times the average step change in the preceding slow-rise region. After this jump, D_{enh} continues to rise but at a much slower rate (+1.4% over the remaining three steps).

The preset criterion for a positive discovery is satisfied: the default diagnosis is blind to the restructuring while the relation-structure probe captures a sudden, large-amplitude reorganization. We term this phenomenon *silent discordance*, located at $\mu \approx 1.5$.

IV. EXPERIMENT 6: TAIL-SLOPE DIAGNOSIS

A. Data

Table II reports the full dataset, now including the tail-slope probe S_{tail} and the step-to-step change ΔS_{tail} .

B. Application of the preset criterion

The preset criterion asks: is ΔS_{tail} across the step $\mu = 1.43 \rightarrow 1.57$ larger than the maximum ΔS_{tail} observed in the seven preceding intervals ($\mu = 0.57 \rightarrow 1.43$)? The maximum in the reference window is 0.0290 (attained at the step $1.29 \rightarrow 1.43$). The value at the critical step is $0.0313 > 0.0290$. Formally, the binary criterion yields a positive result.

TABLE II. Experiment 6: Tail-slope diagnosis added to the silent discordance scan.

μ	D_{def}	D_{enh}	S_{tail}	ΔS_{tail}
0.00	0.0656	0.333121	-3.9513	—
0.14	0.0659	0.331922	-3.9538	0.0025
0.29	0.0662	0.330883	-3.9585	0.0047
0.43	0.0663	0.330117	-3.9656	0.0071
0.57	0.0662	0.329742	-3.9756	0.0100
0.71	0.0659	0.329854	-3.9887	0.0131
0.86	0.0652	0.330492	-4.0051	0.0164
1.00	0.0643	0.331624	-4.0250	0.0199
1.14	0.0632	0.333146	-4.0481	0.0231
1.29	0.0620	0.334912	-4.0743	0.0262
1.43	0.0608	0.336767	-4.1033	0.0290
1.57	0.0596	0.360240	-4.1346	0.0313
1.71	0.0584	0.362081	-4.1678	0.0332
1.86	0.0574	0.363723	-4.2025	0.0347
2.00	0.0566	0.365145	-4.2383	0.0358

C. Critical re-examination

However, inspection of the full ΔS_{tail} column reveals a clear trend: the step-to-step change increases continuously from 0.0025 at the first interval to 0.0358 at the last. There is no isolated peak at $\mu = 1.5$; the change at the critical step merely continues a pre-existing acceleration. The tail slope does not follow the “one-time jump” pattern exhibited by the spectral head; instead, it accelerates across the entire scan range and does not stabilize within $\mu \leq 2.0$.

Thus, the formal positive result is an artifact of a binary criterion that was designed for an abrupt, isolated jump. The more faithful description is that *the entanglement spectrum head and tail decouple at the silent discordance point*:

- The head (standard deviation) undergoes a one-time reorganization at $\mu \approx 1.5$ and then stabilizes at a new plateau.
- The tail (slope) begins a sustained acceleration at the same μ , becoming progressively steeper without reaching a new equilibrium.

D. Physical interpretation

The head of the entanglement spectrum encodes the dominant, large-weight correlation modes; its sudden jump indicates a competitive re-shuffling among these leading modes. The tail encodes the weak, small-weight correlations that form the “periphery” of the relation network; its continuous acceleration suggests that once the leading modes have reorganized, a slow, persistent extrusion of the weakest correlations sets in. If we were to extend the scan to larger μ , we would expect the tail to continue steepening, possibly until only two or three eigenvalues retain any significant weight.

V. METHODOLOGICAL LESSONS

The failure of the preset binary criterion in Experiment 6 is instructive. It shows that rigid “positive/negative” decision rules, while essential for preventing the tool-rationality paradox (i.e., post-hoc adjustment of standards), can be too coarse to capture richer physical structures. When data exhibit a pattern that lies outside the originally envisioned dichotomy, the honest course is to report the failure of the criterion, describe the actual observed structure, and propose new, more refined diagnostic frameworks. This is precisely the “diagnose, do not prescribe” principle: the experiment diagnoses not only the system, but also the adequacy of the diagnostic tools themselves.

VI. CONCLUSION

We have presented a detailed two-stage numerical diagnosis of silent discordance in the PXP model at $L = 18$. Experiment 5 provides a clean, reproducible signature: the standard deviation of the entanglement spectrum jumps by +7% at $\mu \approx 1.5$, while the mean correlation remains smooth. Experiment 6 reveals that this restructuring is accompanied by a head-tail decoupling: the spectral head reorganizes once, while the tail begins a continuous acceleration. The preset binary success criterion for Experiment 6 is formally satisfied but is shown to be conceptually inadequate; the true finding is the more nuanced decoupling phenomenon. All data and analysis code are available online [3].

ACKNOWLEDGMENTS

All calculations were performed on a personal computer using NumPy, SciPy, and matplotlib.

[1] G. Li, “Crystal Vein Philosophy: Theoretical Origin, Engineering Practice, and Meta-Reflection,” (2026).

[2] G. Li, “Searching for Relation Before Entity: A Systematic Quantum Many-Body Numerical Experiment,” (2026).

- [3] G. Li, R. Drahos, and M. Sultanov, *edrn-dmrg-verification*, GitHub repository (2026).